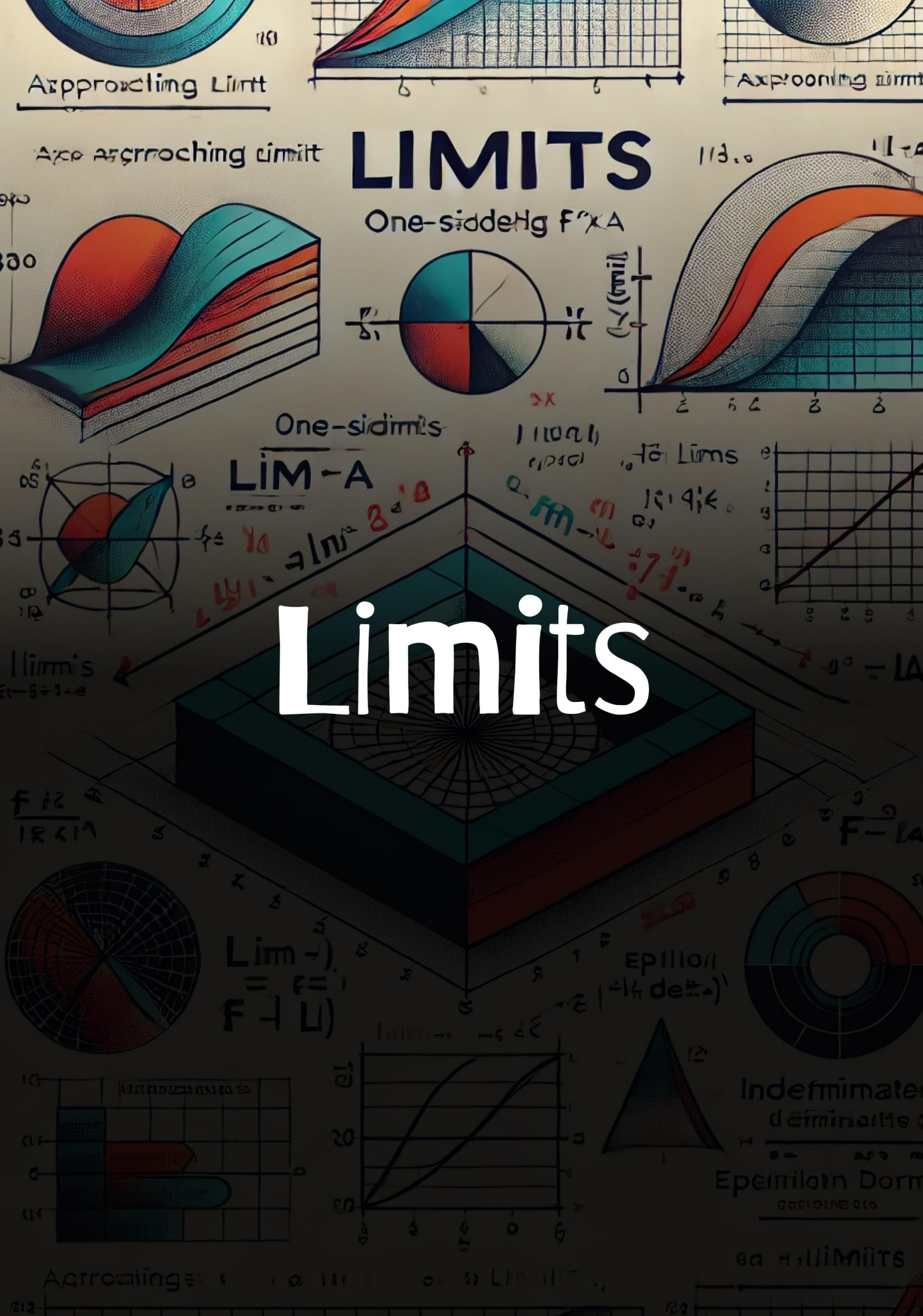
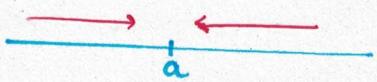

LIMITS
Limits

LIMITS

If x tends to 'a' from left side, we write $x \to a^{\ominus}$

If x tends to 'a' from right side, we write $x \to a^{\oplus}$

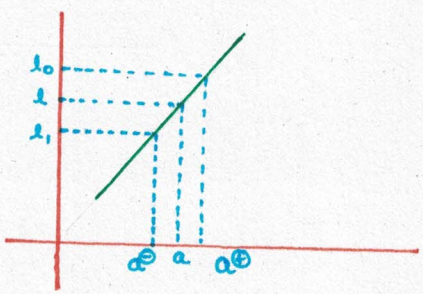


$x \to a$ means x approaches 'a' but not equal to 'a'.

> Even if the value of limit at $x \to a$, i.e. $f(a) \neq L$, then also if $f(a^-) = f(a^+)$, the limit exists. Even the function is not required to be defined at $x = a$

$a^{\ominus}$ → Left hand limit

$a^{\oplus}$ → right hand limit



$\lim_{x \to a} f(x) = l$

When 'x' approaches 'a' then $f(x)$ approaches to a 'unique' no. 'l'.

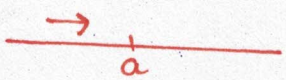
LEFT HAND LIMIT

When x approaches 'a' from left side i.e. $x \to a^{\ominus}$

$\lim_{x \to a^{\ominus}} f(x) = l_1$

$a^{\ominus} = a - h$

$\lim_{h \to 0} f(a-h) = l_1$



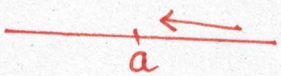
RIGHT HAND LIMIT

When 'x' approaches 'a' from right side i.e. $x \to a^{\oplus}$

$\lim_{x \to a^{\oplus}} f(x) = l_2$

$a^{\oplus} = a + h$

$\lim_{h \to 0} f(a+h) = l_2$



NOTE For a limit to exist at $x = a$,
$\lim_{x \to a} f(x)$ exist if LHL = RHL i.e. $l_1 = l_2$

Ques: Check if limit exists or not?

(i) $f(x) = \frac{x^2-4}{x-2}$, $x \neq 2$, $= 0$, $x = 2$

→ $f(x) = x+2$, $x \neq 2$

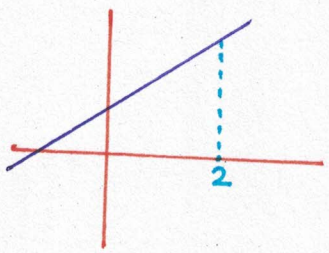
$= 0$, $x = 2$

at $x = 1.99$, $y = 3.99 \to 4$

$x = 2.01$, $y = 4.01 \to 4$

Hence LHL = RHL

∴ Limit exists



(ii) $f(x) = \begin{cases} 2x-1 & , \ 0 \leq x < 1 \\ x^2 & , \ 1 \leq x < 4 \end{cases}$

→ $y = 2x-1$, $x \in [0,1)$

at $x = 0.01$ at $x = 0$, $y = -1$

$y = 0.2 - 1 = -0.98 \to -1$

$y = x^2$, $x \in [1,4]$

at $x = 1$, $y = 1$

at $x = 1.001$, $y \to 1$

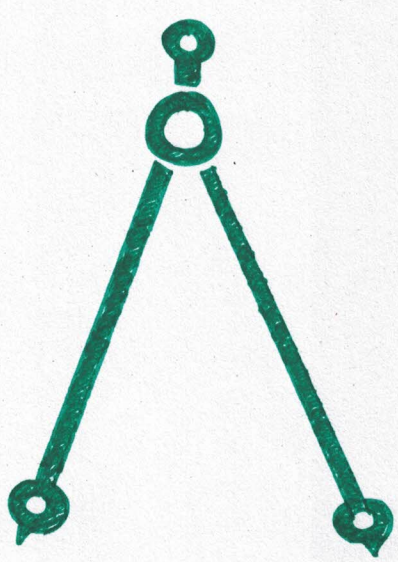
We have to define $f(x)$ for 1

$\lim_{x \to 1} (2x-1) \Rightarrow 2-1 = 1$

$\lim_{x \to 1} (x^2) \Rightarrow 1^2 = 1$

AS LHL = RHL

∴ Limit exists



(iii) $f(x) = \sqrt{x}$

$\lim_{x \to 0} \sqrt{x}$

→ Domain E $[0, \infty)$ ($0^{\ominus}$ is not defined)

at $x = 0$, $y = 0$

at $x = 0.001$, $y \to 0$

∴ Limit exists and limit = 0

(iv) $\lim_{x \to 1} [\sin^{-1} x]$

→ Domain $x \in [-1,1]$

at $x = 1$, $y = \left[\frac{\pi}{2}\right] = 1$

at $x \to 1$, $y \to \left[\frac{\pi}{2}\right] = 1$

Limit exists and is 1.

[$1^{\oplus}$ does not exist]

(v) $f(x) = \begin{cases} 1, & x \in Q \\ 1-x, & x \notin Q \end{cases}$ Find $\lim_{x \to 0} f(x)$

→ At $x \to 0^{\oplus}$, $f(x) = 1$

↓

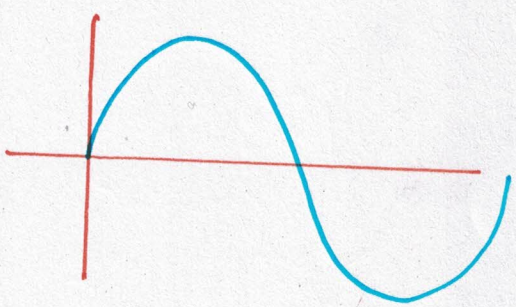
rational

$\lim_{x \to 0^{\oplus}} f(1-x) = 1$

$\lim_{x \to 0} f(x) = 1$

(v) $\lim_{x \to \frac{5\pi}{4}} [\sin x + \cos x]$

→ $\left[\left(\frac{\sin x}{\sqrt{2}} + \frac{\cos x}{\sqrt{2}}\right)\sqrt{2}\right] \Rightarrow \left[\sin\left(x + \frac{\pi}{4}\right)\sqrt{2}\right]$



At $x \to \frac{5\pi}{4}$, $x + \frac{\pi}{4} \to \frac{3\pi}{2}$

$\sin\left[x + \frac{\pi}{4}\right] \to \sin\left(\frac{3\pi}{2}\right)$

For $\frac{3\pi}{2} + \Delta\theta$, $\sin\left(\frac{3\pi}{2} + \Delta\theta\right) \to -1$

For $\frac{3\pi}{2} - \Delta\theta$, $\sin\left(\frac{3\pi}{2} - \Delta\theta\right) \to -1$

$\lim_{x \to \frac{5\pi}{4}} [\sin x + \cos x] = 2$

INDETERMINANT FORM OF LIMIT

$\lim_{x \to 0} \frac{f(x) \to 0}{g(x) \to 0}$ indeterminant

Let $\lim_{x \to 0} \frac{f(x)}{g(x)}$ = determinant = l

$\frac{\to 0}{\to 0} = l$

$\Rightarrow \to 0 = l \times (\to 0)$

But l can be anything here but for limit to exist l should be unique. Hence it is indeterminant.

① $\frac{\to 0}{\to 0}$ form is indeterminant

* If $f(x) \to \infty$, $g(x) \to \infty$

$\frac{\to\infty}{\to\infty} = l \quad \Rightarrow \to\infty = lX \to \infty$

l is not unique

∴ ② $\frac{\to\infty}{\to\infty}$ form is indeterminant

* $\lim_{x\to 0} f(x) - g(x)$

Let $f(x) \to \infty$, $g(x) \to \infty$

$(\to\infty) - (\to\infty)$

Let $(\to\infty) - (\to\infty) = l$

$\to\infty = l + (\to\infty)$

l is not unique

③ $(\to\infty) - (\to\infty)$ form is indeterminant

$\lim_{x\to 0} [f(x)]^{g(x)}$

Let $\lim_{x\to 0} f(x) \to 1$

$\lim_{x\to 0} g(x) \to \infty$

$(\to 1)^{\to\infty}$

For 0.99, $(0.99)^{\to\infty} \to 0$

For 1.01, $(1.01)^{\to\infty} \to 1$

So, we don't get a unique value

∴ ④ $(\to 1)^{\to\infty}$ form is indeterminant

* $\lim_{x\to a} [f(x)]^{g(x)}$

Let $\left[\lim_{x\to a} f(x) \to 0 \quad \lim_{x\to a} g(x) \to 0\right]$

For ⊕, ⊖ we will not get unique value

⑤ $(\to 0)^{\to 0}$ form is indeterminant

SOME PROPERTIES OF LIMITS

1. $\lim_{x\to a} (K_1 f(x) \pm K_2 g(x)) \Rightarrow K_1 \lim_{x\to a} f(x) \pm K_2 \lim_{x\to a} g(x)$
2. $\lim_{x\to a} (f(x)\, g(x)) = \lim_{x\to a} f(x) \lim_{x\to a} g(x)$
3. $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$; $\lim_{x\to a} g(x) \neq 0$
4. $\lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right)$
5. $\lim_{x\to 0} \frac{\sin x}{x} = 1$
6. $\lim_{x\to 0} \frac{\tan x}{x} = 1$

Ques: Find the value of $\lim_{x\to 0^{\oplus}} \left[\frac{\sin x}{x}\right]$

Sol: $x > 0$

$x > \sin x$

$\frac{\sin x}{x} < 1 = 1^{\ominus}$

$\left[\lim_{x\to 0^{\oplus}} \frac{\sin x}{x}\right] = [1^{\ominus}] = 0$

$a < 0$,

$x < \sin x$

$\frac{\sin x}{x} > 1 = 1^{\oplus}$

$\left[\lim_{x\to 0^{\ominus}} \frac{\sin x}{x}\right] = [1^{\oplus}] = 1$

BOUNDED FUNCTION

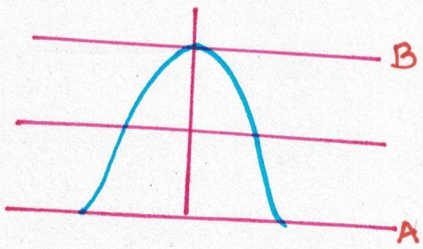
[If $f: D \longrightarrow$ Range, $y = f(x)$]

$f(x)$ has a lower bound if there exist a number 'A' such that $f(x) \geq A \ \forall\ x \in D$

$f(x)$ has a upper bound if there exist a number 'B' such that $f(x) \leq B$, $\forall\ x \in D$

$f(x)$ has a lower bound & upper bound both $\forall\ x \in D$, then $f(x)$ is called a bounded function i.e. $A \leq f(x) \leq B\ \forall\ x \in D$

Graph of bounded function $f(x)$ lies b/w $y=A$ & $y=B$



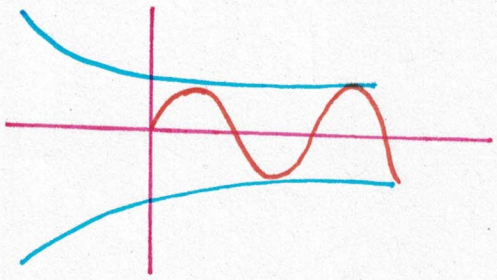
NOTE $\lim\limits_{x\to a} f(x)\,g(x)$ if $g(x)$ is a bounded function $\forall\ x \in D$ (Domain of $g(x)$) and $\lim\limits_{x\to a} f(x)$ is infinitisimally small quantity then $\lim\limits_{x\to a} f(x)\,g(x) = 0$

Ques: $\lim\limits_{x\to\infty} e^{-x}\sin x$

Sol: $\sin x \to$ bounded

$x \to \infty$, $e^{-x} \to 0$

$\lim\limits_{x\to\infty} e^{-x}\sin x = 0$

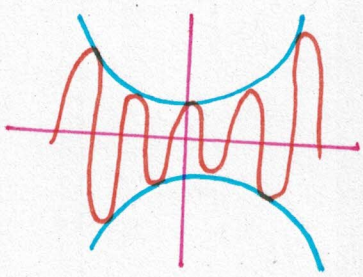


Ques: $\lim\limits_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$

Sol: $\sin\left(\frac{1}{x}\right) \to$ bounded

$x \to 0$, $x^2 \to 0$

$\lim\limits_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$



NOTE $\lim\limits_{x\to a} f(x)$ if one side of limit does not lie inside the domain of $f(x)$ then limiting value of $f(x)$ is same as the limiting value of other side of the limit

Eg. $\lim\limits_{x\to 0} \sqrt{x}$, $\lim\limits_{x\to 1} \sin^{-1}(x)$, $\lim\limits_{x\to 3} \dfrac{[x^2]-2[x]-3}{[x^2]-4[x]+3}$

Ques: $\lim\limits_{x\to 3} \dfrac{[x^2]-2[x]-3}{[x^2]-4[x]+3}$

Sol: $x = 3^{\ominus}$, $\dfrac{[8\ldots] - 2[2\ldots] - 3}{[8\ldots] - 4[2\ldots] + 3}$ $\quad ([x]-1)([x]-3) \neq 0$

$[x] \neq 1, \quad [x] \neq 3$

$\Rightarrow \dfrac{8-4-3}{8-8+3} = 1/3$

$x = 3^{\oplus}$ does not exist

Ques: If $\lim\limits_{x\to 2} \dfrac{x^3 + 3ax + 2}{x^2 - x - 2}$ exists. Find a

Sol:

$\lim\limits_{x\to 2}$ $x^2 - x - 2 \to 0$

As limit exists $x-2$ must be a factor in numerator also, so at $x = 2$ numerator is -

$8 + 6a + 2 = 0$

$a = -\dfrac{5}{3}$

Ques: If $f(x) = \begin{cases} \dfrac{\sqrt{x^2 - 6x + 9}}{x-3}, & x \neq 3 \\ 0, & x = 3 \end{cases}$ Find $\lim\limits_{x\to 3} f(x)$

Sol: $f(x) = \dfrac{\sqrt{x^2 - 6x + 9}}{x-3}$

$= \dfrac{\sqrt{(x-3)^2}}{x-3}$

$= \dfrac{|x-3|}{x-3}$

$3^{\oplus}$, $f(x) = \dfrac{|3+h-3|}{3+h-3} = \dfrac{h}{h} = 1$

$3^{\ominus}$, $f(x) = \dfrac{|3-h-3|}{3-h-3} = \dfrac{-h}{h} = -1$

Limit does not exists as LHL $\neq$ RHL

Ques: If $f(x) = \begin{cases} \dfrac{\sin[x]}{[x]}, & [x] \neq 0 \\ 0, & [x] = 0 \end{cases}$ find $\lim\limits_{x\to 0} f(x)$

Sol:

0^{+}, $f(x) = \dfrac{\sin[0+h]}{[0+h]} = \dfrac{\sin[h]}{[h]} = \dfrac{\sin 0}{0} = 0$ } given in question

$0^{\ominus}$, $f(x) = \dfrac{\sin[0-h]}{[0-h]} = \dfrac{\sin(-1)}{-1} = \sin 1$

Limit does not exist

Ques: Find $\lim\limits_{x\to 0}\left(\tan^{-1}\dfrac{1}{x} + \cot^{-1}\dfrac{1}{x^2}\right)$

Sol: $x = 0^{+}$, $\tan^{-1}\dfrac{1}{(0+h)} + \cot^{-1}\dfrac{1}{(0+h)^2}$

$\Rightarrow \tan^{-1}\dfrac{1}{h} + \cot^{-1}\dfrac{1}{h^2}$

$x = 0^{-}$, $\tan^{-1}\dfrac{1}{(0-h)} + \cot^{-1}\dfrac{1}{(0-h)^2}$

$\Rightarrow -\tan^{-1}\dfrac{1}{h} + \cot^{-1}\dfrac{1}{h^2}$

Limit does not exist

Ques: $\lim\limits_{x\to 1} x\,\text{sgn}(x-1)$ $\begin{cases} \dfrac{|x|}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$

Sol: at $x = 1^{\oplus}$, $(1+h)\,\text{sgn}(1+h-1)$

$\Rightarrow (1+h)\,\text{sgn}\,h$

$\Rightarrow (1+h)\dfrac{|h|}{h} \Rightarrow (1+h) \times \oplus$

at $x = 1^{\ominus}$, $(1-h)\,\text{sgn}(1-h-1)$

$\Rightarrow (1-h)\,\text{sgn}(-h)$

$\Rightarrow (1-h)\dfrac{|-h|}{-h}$

$\Rightarrow (1-h) \times \ominus$

Limit does not exist

Ques: $\lim\limits_{x\to 2}\{x\}$

Sol: $x = 2^{\oplus}$, $\{2+h\} \Rightarrow h = 2+h-[2+h]$

$\Rightarrow 2+h-2 = h$

$x = 2^{\ominus}$, $\{2-h\} \Rightarrow 2-h-[2-h] \Rightarrow 2-h-1 = 1-h$

Limit does not exist

Ques: If $\lim\limits_{x\to 0}\dfrac{f(x)}{x^2} = 2$ then find ① $\lim\limits_{x\to 0}[f(x)]$ ② $\lim\limits_{x\to 0}\left[\dfrac{f(x)}{x}\right]$

Sol: I] $\lim_{x\to 0} \frac{f(x)}{x^2} = 2$

$0^{\oplus}$, $\frac{f(0+h)}{(0+h)^2} = 2 \Rightarrow f(h) = 2h^2$

$0^{\ominus}$, $\frac{f(0-h)}{(0-h)^2} = 2 \Rightarrow f(-h) = 2h^2$

(i) $\lim_{x\to 0}[f(x)] \Rightarrow x = 0^{\oplus}, [f(0+h)] \Rightarrow [f(-h)] = 2h^2$

$= 0$ Ans.

(ii) $\lim_{x\to 0}\left[\frac{f(x)}{x}\right] \Rightarrow x = 0^{\oplus}, \left[\frac{f(h)}{h}\right] = \frac{2h^2}{h} = [2h] = 0$

$x = 0^{\ominus}, \left[\frac{f(-h)}{-h}\right] = \left[\frac{2h^2}{-h}\right] = [-2h] = -1$

Limit does not exists

II] (i) $\lim_{x\to 0}[f(x)] = \lim_{x\to 0}\left[\frac{f(x)}{x^2}\cdot x^2\right] = [2x^2] = 0$

(ii) $\lim_{x\to 0}\left[\frac{f(x)}{x}\right] = \lim_{x\to 0}\left[\frac{f(x)}{x^2}\cdot x\right] = [2x] = 0$

$0^{\ominus}, \left[\frac{f(x)}{x^2}\cdot x\right] = [-2x] = -1$

Limit does not exist

$\lim_{x\to a} \frac{x^n - a^n}{x-a} = n a^{n-1}$

Proof: $\frac{x^n - a^n}{x-a} \Rightarrow \frac{\left(\left(\frac{x}{a}\right)^n - 1\right)a^n}{\left(\frac{x}{a} - 1\right)a} \Rightarrow a^{n-1}\frac{(r^n-1)}{(r-1)}$ } sum of a GP.

$\Rightarrow a^{n-1}(1 + r + r^2 + \dots r^{n-1})$

$\Rightarrow a^{n-1}\left(1 + \frac{x}{a} + \left(\frac{x}{a}\right)^2 + \dots \left(\frac{x}{a}\right)^{n-1}\right)$

$\Rightarrow n a^{n-1}$

$x\to a$

Ques: $\lim_{x\to 5} \frac{x^3 - 125}{x^2 - 25}$

Sol: $\lim_{x\to 5} \frac{(x-5)(x^2+25+5x)}{(x-5)(x+5)}$

$\Rightarrow \lim_{x\to 5} \frac{x^2+5x+25}{x+5}$

$\Rightarrow \frac{25+25+25}{10} = 7.5$

Ques: $\lim_{x\to 1} \frac{x^{p+1} - (p+1)x + p}{(x-1)^2}$, $p \in N$

Sol: $\lim_{x\to 1} \frac{x^{p+1} - (p+1)x + p}{(x-1)^2} \Rightarrow \lim_{x\to 1} \frac{x(x^p-1) - p(x-1)}{(x-1)^2}$

$\Rightarrow \lim_{x\to 1} \frac{\frac{x(x^p-1)}{x-1} - p}{(x-1)}$

$\Rightarrow \lim_{x\to 1} \frac{(x+x^2+\ldots x^p) - p}{(x-1)}$

$\Rightarrow \lim_{x\to 1} \frac{(x+x^2+\ldots+x^p) - (1+1+\ldots p)}{(x-1)}$

$\Rightarrow \lim_{x\to 1} \frac{(x-1)}{x-1} + \frac{(x^2-1)}{x-1} + \ldots \frac{(x^p-1^p)}{x-1}$

$\Rightarrow 1+2+3+\ldots p \quad \left[\frac{x^n-a^n}{x-a} = na^{n-1}\right]$

$\Rightarrow \frac{p(p+1)}{2}$

Ques: $\lim_{x\to 1} \left(\frac{4-\sqrt{5x+11}}{2-\sqrt{3x+1}}\right)$

Sol: $\lim_{x\to 1} \left(\frac{4-\sqrt{5x+11}}{1-\sqrt{3x+1}}\right)\left(\frac{4+\sqrt{5x+11}}{2+\sqrt{3x+1}}\right)\left(\frac{2+\sqrt{3x+1}}{4+\sqrt{5x+11}}\right)$

$\lim_{x\to 1} \left(\frac{18-5x-11}{4-3x-1}\right)\left(\frac{2+\sqrt{3x+1}}{4+\sqrt{5x+11}}\right)$

$\lim_{x\to 1} \left(\frac{5-5x}{3-3x}\right)\left(\frac{2+\sqrt{3x+1}}{4+\sqrt{5x+11}}\right)$

$\lim_{x\to 1} \frac{5}{3} \times \frac{2+\sqrt{3x+1}}{4+\sqrt{5x+11}}$

$\Rightarrow \frac{5}{3} \times \frac{4}{8} = \frac{5}{6}$ Ans.

Ques: $\lim_{x\to 1} \sin^{-1}\left(\frac{1-\sqrt{x}}{1-x}\right)$

Sol: $\sin^{-1}\left(\frac{1-\sqrt{x}}{(1-\sqrt{x})(1+\sqrt{x})}\right)$

$= \sin^{-1}\left(\frac{1}{1+\sqrt{x}}\right) \Rightarrow \lim_{x\to 1} \sin^{-1}\left(\frac{1}{2}\right) \Rightarrow \pi/6$

Ques: $\lim_{x\to 0}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{x}\right)$

Sol: $\frac{\sqrt{1+x}-\sqrt{1-x}}{x} \times \left(\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$

$= \frac{1+x-(1-x)}{x(\sqrt{1+x}-\sqrt{1-x})}$

$= \frac{2x}{x(\sqrt{1+x}+\sqrt{1-x})}$

$= \frac{2}{\sqrt{1+x}+\sqrt{1-x}}$

$\lim_{x\to 0} \frac{2}{2} = 1$ Ans.

Ques: $\lim_{x\to 0}\left(\frac{\sqrt{1-\cos x}}{x}\right)$

Sol. $\frac{\sqrt{1-\cos x}}{x}$, 0^+, $\frac{\sqrt{1-\cos h}}{h}$

$0^{\ominus}$, $\frac{\sqrt{1-\cos h}}{-h}$

LHL ≠ RHL

Limit does not limit

LIMITS USING SERIES EXPANSION OF FUNCTION

MACLAREN'S THEOREM

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots\dots$$

eg. $f(x) = e^x$ $\quad f'(x) = e^{+x} = f''(x) = e^x$

$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\dots$

(i) $a^x = f(x)$

$a^x = e^{x \ln a}$

$$f(x) = 1 + x\ln x + \frac{x^2 \ln^2 x}{2!} + \frac{x^3 \ln^3 x}{3!} + \dots\dots$$

(ii) $f(x) = \ln(1+x)$

$f(x) = 0 + x\left(\frac{1}{1+x}\right) + x^2\left(\frac{1}{(1+x)^2}\right) + \dots\dots$

$$f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots\dots$$

Ques: $\dfrac{x\tan 2x - 2x\tan x}{x^2 - \sin^2 x}$ $\underset{x\to 0}{Lt}$?

Sol:

$$\frac{\dfrac{2x\tan x}{1-\tan^2 x} - 2x\tan x}{x^2 - \sin^2 x} = \frac{2\tan x\left(\dfrac{1}{1-\tan^2 x} - 1\right)}{2\left(\dfrac{x+\sin x}{x}\right)(x-\sin x)}$$

$$= \frac{2\tan^3 x}{(1-\tan^2 x)(x-\sin x)\times 2}$$

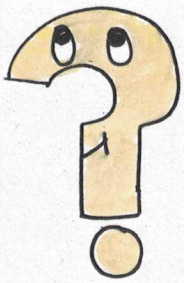
$$= \frac{\dfrac{\tan^3 x}{x^3}\times x^3}{x-\sin x}$$

$$= \frac{x^3}{x-\sin x} = \frac{3x^2}{1-\cos x}$$

$$= \frac{6x}{0+\cos x} = \frac{6}{\cos x} = 6$$

Series Expansion

1. $e^x = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \dots\dots$
2. $a^x = 1 + x\ln a + \dfrac{x^2\ln^2 a}{2!} + \dfrac{x^3\ln^3 a}{3!} + \dots$
3. $\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dots\dots$
4. $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \dfrac{x^7}{7!} + \dots$
5. $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} \dots$
6. $\tan x = x + \dfrac{x^3}{3} + \dfrac{2x^5}{15} + \dots\dots$
7. $\tan^{-1} x = x - \dfrac{x^3}{3} + \dfrac{x^5}{5} - \dfrac{x^7}{7} + \dots$
8. $\sin^{-1} x = x + \dfrac{1^2}{3!}x^3 + \dfrac{1^2\cdot 3^2\cdot x^5}{5!} + \dfrac{1^2\cdot 3^2\cdot 5^2 x^7}{7!} + \dots\dots$
9. $(1+x)^n = 1 + nx + \dfrac{n(n-1)x^2}{2!} + \dots\dots \quad \{-1 < x < 1\}$
10. $(1+x)^{1/x} = e\left(1 - \dfrac{x}{2} + \dfrac{11}{24}x^2 + \dots\right)$



Ques: $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$

Sol: $$\frac{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) - 1 - x}{x^2}$$

$$= \frac{1}{2!} + (\text{Term with } x)$$

$$= \frac{1}{2} \qquad (x \to 0)$$

Ques: $\lim_{x\to 0} \dfrac{\tan x - \sin x}{x^3}$

Sol: $$\frac{\left(a + \frac{x^3}{3} + \frac{2x^2}{15} + \dots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots\right)}{x^3}$$

$$\Rightarrow \left(\frac{1}{3} + \frac{1}{3!}\right) + (\text{Term with } x)$$

As $x \to 0$

$$\Rightarrow \frac{1}{3} + \frac{1}{3!} \qquad \Rightarrow \frac{1}{3} + \frac{1}{6} = 1/2 \text{ Ans.}$$

Ques: $\lim_{x\to 0} \dfrac{e^x + \ln\left(\frac{1-x}{e}\right)}{\tan x - x}$

Sol: $$\frac{e^x - \frac{e}{1-x} \times \frac{1}{e}}{\sec^2 x - 1}$$

$$= \frac{e^x + \ln(1-x) - 1}{\tan x - x}$$

$$= \frac{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) + \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \dots\right)}{\left(x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots\right) - x}$$

$$= \frac{\left(\frac{x^3}{3!} + \dots\right) - \left(\frac{x^3}{3} + \dots\right)}{\frac{x^3}{3} + \frac{2x^5}{15} + \dots}$$

$$= \frac{x^3\left(\frac{1}{3!} - \frac{1}{3}\right) + (\text{Terms with } x)}{\frac{x^3}{3} + \frac{2x^5}{15} + \dots}$$

$$= \frac{x^3\left(\frac{1}{3!} - \frac{1}{3}\right) + (\text{Terms with } x)}{\frac{x^3}{3}\left(1 + \frac{2x^2}{5} + \dots\right)}$$

$$= \frac{1}{3!} - \frac{1}{3}(1 + \dots) \Big/ \frac{1}{3}(1 + \dots)$$

$$= \frac{\frac{1}{3!} - \frac{1}{3}}{1/3}$$

$$= \frac{\frac{1}{6} - \frac{1}{3}}{1/3} = \frac{1}{2} - 1 = -\frac{1}{2} \quad \text{Ans.}$$

Ques: $\lim_{x \to 0} \dfrac{5\sin x - 7\sin 2x + 3\sin 3x}{x^2 \sin x}$

Sol:

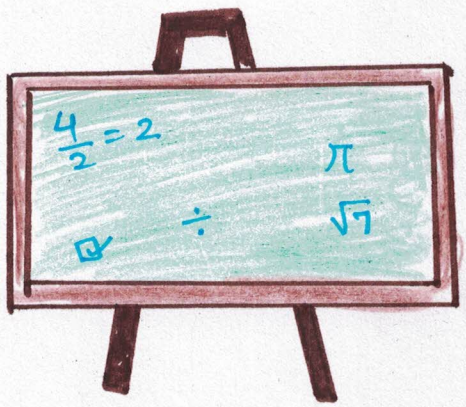
$$\frac{5\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}\right) - 7\left(2x - \frac{8x^3}{3!} + \frac{32x^5}{5!} - \dots\right) + 3\left(3x - \frac{27x^3}{3!} + \frac{243x^5}{5!}\right)}{x^2\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}\right)}$$

$$= \frac{\left(-\frac{5x^3}{3!} + \frac{5x^5}{5!} - \frac{5x^7}{7!} + \dots\right) - \left(-\frac{56x^3}{3!} + \frac{224x^5}{5!} + \dots\right) + \left(-\frac{81x^3}{3!} + \frac{729x^5}{5!}\right)}{x^3\left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} \dots\right)}$$

$$= \frac{-\frac{30}{3!}x^3 + \frac{5x^5}{5!} - \frac{224x^5}{5!} + \frac{729}{5!}x^5 + \dots}{x^3\left(1 - \frac{x^2}{3!} + \dots\right)}$$

$$= \frac{x^3\left(-\frac{30}{3!} + \frac{5x^2}{5!} + \dots\right)}{x^3\left(1 - \frac{x^2}{3} + \dots\right)}$$

$$= \frac{-30}{3!} = \frac{-30}{6} = -5 \quad \text{Ans.}$$



Ques: $\dfrac{5\sin x - 7(2\sin x \cos x) - 3(3\sin x - 4\sin^3 x)}{x^2 \sin x}$

Sol:

$$\frac{14 - 14\cos x - 12\sin^2 x}{x^2}$$

$$= \frac{14 \times 2\sin^2\frac{x}{2}}{\frac{x^2}{4} \times 4} - 12\,\frac{\sin^2 x}{\frac{x}{2}}$$

$$= \frac{14 \times 2}{4} - 12(1)$$

$$= -5$$

Ques: $\lim_{x \to 0} \dfrac{e^{\sin x} - \sin x - 1}{x^2}$

Sol:

$$\frac{e^{\sin x}\cos x - \cos x}{2x}$$

$$= \left(1 + \sin x + \frac{\sin^2 x}{2!} + \frac{\sin^3 x}{3!} + \dots\right) - 1 - \sin x \Big/ x^2$$

$\Rightarrow \frac{\sin^2 x}{2x^2} + \frac{\sin^3 x}{3! x^2} + \dots$

$\Rightarrow \frac{1}{2}\left(\frac{\sin x}{x}\right)^2 + \dots$

$\Rightarrow \boxed{\frac{1}{2}}$ Ans.

Ques: $\lim_{x\to 0} \frac{a\cos x + bx\sin x - 5}{x^4}$ exists. Find $a+b$.

Sol: $\frac{a\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right) + bx\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \dots\right) - 5}{x^4}$

$\Rightarrow \frac{a\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots\right) + b\left(x^2 - \frac{x^4}{3!} + \frac{x^6}{6!} - \frac{x^8}{7!}\right) - 5}{x^4}$

$\Rightarrow \frac{(a-5) + x^2\left(b - \frac{a}{2!}\right) + x^4\left(\frac{a}{4!} - \frac{b}{3!}\right) + \dots}{x^4}$

$\Rightarrow a-5=0$ for $\frac{0}{0}$ form which can be then converted

$\boxed{a = 5}$

x^2 coefficient must be zero otherwise $\frac{1}{x^2} \to \infty$ at $x \to 0$

$b = \frac{a}{2!} = \frac{5}{2}$

$\boxed{a+b = \frac{15}{2}}$

$\boxed{\text{Limit} = \frac{5}{4!} - \frac{5}{2 \cdot 3!}}$

Ques: $\lim_{x\to 0} \frac{Ax^2 - B\cos x + Ce^x}{x\sin x} = 2$ then find A, B, C.

Sol: $\frac{Ax^2 - B\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \dots\right) + C\left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots\right)}{x\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots\right)}$

$\Rightarrow \frac{Ax^2 - \left(B - \frac{Bx^2}{2!} + \frac{Bx^4}{4!} \dots\right) + \left(C - Cx + \frac{Cx^2}{2!} - \frac{Cx^3}{3!} + \dots\right)}{x^2\left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \dots\right)}$

$\Rightarrow \frac{(C-B) + x^2\left(A - \frac{B}{2!} + \frac{C}{2!}\right) + x(-C) + \dots}{x^2\left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} + \dots\right)}$

$\boxed{C = 0}$ $\boxed{B = 0}$ $\boxed{A = 2}$

STANDARD LIMITS

1. $\lim_{x\to 0} \frac{\sin x}{x} = 1$ (x is in radian)
2. $\lim_{x\to 0} \frac{\tan x}{x} = 1$
3. $\lim_{x\to 0} \frac{\sin^{-1} x}{x} = 1$
4. $\lim_{x\to 0} \frac{\tan^{-1} x}{x} = 1$
5. $\lim_{x\to 0} \frac{a^x - 1}{x} = \log_e a$
6. $\lim_{x\to 0} \frac{e^x - 1}{x} = \log_e e = 1$
7. $\lim_{x\to 0} \frac{\log(1+x)}{x} = 1$
8. $\lim_{x\to 0} (1+x)^{1/x} = e$

$\Rightarrow \lim_{x\to 0} e^{\frac{1}{x}\log_e(1+x)}$

$\Rightarrow \lim_{x\to 0} e^{\frac{\log_e(1+x)}{x}}$

$\Rightarrow e^1 = e$

Ques: Find the limit of -

① $\lim_{x\to 0} \frac{\tan x}{x}$

$180^\circ \to \pi$

$1^\circ \to \frac{\pi}{180^\circ}$

$\Rightarrow \frac{\tan \frac{\pi x}{180}}{x} = \frac{\tan \frac{\pi}{180} x}{\frac{\pi}{180} x \times \frac{180}{\pi}} \Rightarrow \frac{\pi}{180}$

② $\lim_{x\to 0} \frac{\ln(\cos(\sin x))}{x^2}$

$\Rightarrow \frac{\ln\left(1 + \left(-2\sin^2 \frac{\sin x}{2}\right)\right)}{x^2}$

$\Rightarrow \frac{\ln\left(1 + \left(-2\sin^2 \frac{\sin x}{2}\right)\right) \times \left(-2\sin^2 \frac{\sin x}{2}\right)}{-2\sin^2 \frac{\sin x}{2} \times x^2}$

$\Rightarrow \lim_{x\to 0} \frac{-2\sin^2 \frac{\sin x}{2}}{x^2 \frac{\sin^2 x}{4}} \times \frac{\sin^2 x}{4}$

$\Rightarrow \lim_{x\to 0} \frac{-2}{x^2} \times \frac{\sin^2 x}{4} = -\frac{4}{2}\left(\frac{\sin x}{x}\right)^2 = -\frac{1}{2}$

Ques: $\lim_{x \to 2} \frac{x^2-4}{\cos \frac{\pi}{4} x}$

Sol: $2^+, \; 2+h \Rightarrow \frac{(2+h)^2-4}{\cos \frac{\pi}{4}(2+h)}$

$\Rightarrow \frac{(2+h)^2-4}{\cos\left(\frac{\pi}{2}+\frac{\pi h}{4}\right)}$

$\Rightarrow \frac{h^2+4h}{-\sin \frac{\pi}{4} h}$

$\Rightarrow \frac{h(h+4)}{-\sin \frac{\pi}{4} h} = \frac{h+4}{\frac{-\sin \frac{\pi}{4} h}{h \times \pi/4}} \times \pi/4$

$= -h+4 \times \frac{4}{\pi}$

$= \boxed{\frac{-16}{\pi}}$

Ques: $\lim_{x \to 0} \frac{\sin 2x}{\sin 3x}$

Sol: $\frac{\sin 2x \times 3x \times 2x}{2x \times \sin 3x \times 3x} = \frac{2}{3}$

Ques: $\lim_{x \to a} \frac{(a^x - a^a)}{(x-a)}$

Sol: $\frac{a^a (a^{x-a}-1)}{(x-a)} \Rightarrow a^a \ln a$

Ques: $\lim_{x \to 0} \frac{\tan x - \sin x}{x^3}$

Sol: $\frac{\frac{\sin x}{\cos x} - \sin x}{x^3}$

$\Rightarrow \frac{\sin x (1-\cos x)}{\cos x^3}$

$\Rightarrow \frac{\sin x \times 2 \sin^2 \frac{x}{2}}{\cos x \; x^3}$

$\Rightarrow \frac{\sin x}{x} \times 2 \times \frac{\sin^2 \frac{x}{2}}{\frac{x^2}{4} \times 4}$

$\Rightarrow 1 \times 2 \times \frac{1}{4} = \boxed{1/2}$

$\lim_{x\to\infty} \frac{P(x)}{Q(x)}$ where $P(x)$ & $Q(x)$ are polynomial in x

$P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots a_0$

$Q(x) = b_m x^m + b_{m-1}x^{m-1} + b_{m-2}x^{m-2} + \dots b_0$

$\lim_{x\to\infty} \frac{P(x)}{Q(x)} \Rightarrow \lim_{x\to\infty} \frac{a_n x^n + a_{n-1}x^{n-1} + \dots a_0}{b_m x^m + b_{m-1}x^{m-1} + \dots b_0}$

(i) $n = m$, $\frac{a_n}{b_m}$

(ii) $n > m$, ∞

(iii) $n < m$, 0

Ques: $\lim_{x\to\infty} \frac{2x^2+3x+1}{x^2-7x+5}$

Sol: As $n=m$, $\lim_{x\to\infty} \Rightarrow \frac{2}{1} = 2$

Ques: $\lim_{x\to\infty} \frac{3x^2+7x+4}{x^3+3x^2+7x+8}$

Sol: As $n<m$, $\lim_{x\to\infty} = 0$

Ques: $\lim_{x\to\infty} (\sqrt{1+2x} - \sqrt{1-2x})$

Sol: $\frac{(\sqrt{1+2x} - \sqrt{1-2x})(\sqrt{1+2x} + \sqrt{1-2x})}{(\sqrt{1+2x} + \sqrt{1-2x})}$

$= \frac{1+2x-(1-2x)}{\sqrt{1+2x} + \sqrt{1-2x}}$

$= \frac{4x}{\sqrt{1+2x} + \sqrt{1-2x}}$

$= \frac{4x}{\sqrt{x}\left[\sqrt{\frac{1}{x}+2} - \sqrt{\frac{1}{x}-2}\right]}$

$= \frac{4\sqrt{x}}{\left(\sqrt{\frac{1}{x}+2} - \sqrt{\frac{1}{x}-2}\right)}$

At $x \to \infty$, $\sqrt{\frac{1}{x}-2} \to \sqrt{-2}$

$\therefore$ Limit does not exists

Ques: $\lim_{x\to\infty} \left(\sqrt{3x+2} - \sqrt{2x+3}\right)$

Sol: $$\frac{\left(\sqrt{3x+2} - \sqrt{2x+3}\right)\left(\sqrt{3x+2} + \sqrt{2x+3}\right)}{\left(\sqrt{3x+2} + \sqrt{2x+3}\right)}$$

$$\Rightarrow \frac{3x+2-(2x+3)}{\sqrt{3x+2}+\sqrt{2x+3}} \Rightarrow \frac{x-1}{\sqrt{3x+2}+\sqrt{2x+3}}$$

$$\Rightarrow \frac{x-1}{\sqrt{x}\left[\sqrt{3+\frac{2}{x}}+\sqrt{2+\frac{3}{x}}\right]} \Rightarrow \frac{x-1}{\sqrt{x}(\sqrt{3}+\sqrt{2})}$$

$\Rightarrow \infty$

Ques: $\lim_{x\to\infty} \dfrac{\sqrt{3x+4} - \sqrt{x+4}}{\sqrt{x+7}}$

Sol: $$\frac{(3x+4)-(x+4)}{\left(\sqrt{3x+4}+\sqrt{x+4}\right)\left(\sqrt{x+7}\right)} = \frac{2x}{\left(\sqrt{3x+4}+\sqrt{x+4}\right)\left(\sqrt{x+7}\right)}$$

$$\Rightarrow \frac{2x}{\sqrt{x}\sqrt{x}\left[\sqrt{3+\frac{4}{x}}+\sqrt{1+\frac{4}{x}}\right]\left[\sqrt{1+\frac{7}{x}}\right.}$$

$$\Rightarrow \frac{2}{(\sqrt{3}+1)(1)} \Rightarrow \frac{2}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$\Rightarrow \boxed{\sqrt{3}-1 \quad \text{Ans.}}$

Ques: $\lim_{x\to\infty} \tan^{-1}\left(\frac{1}{x}\right)$

Sol: $\to 0$

Ques: $\lim_{x\to\infty} \tan^{-1}\frac{1}{x} + \cot^{-1}\frac{1}{x^2}$

Sol: $\tan^{-1}\frac{1}{\infty} + \cot^{-1}\frac{1}{\infty}$

$0 + \pi/2 = \pi/2$

Ques: $\lim_{x\to\infty} \sin\left(\frac{1}{x}\right)$

Sol: $\to 0$

Ques: $\lim_{x\to\infty} \sin x$

Sol: We don't know the exact value of the sin so will give. So, limits do not exist.

1^{∞} Form

$$\lim_{x\to a} f(x)^{g(x)}$$

$$\lim_{x\to a} f(x) \to 1 \quad \& \quad g(x) \to \infty$$

$$\therefore f(x)^{g(x)} = \left(1+f(x)-1\right)^{\frac{1}{f(x)-1}\times f(x-1)\,g(x)}$$

Let $f(x)-1 = X$

$$\lim_{x\to a} (1+X)^{\frac{1}{X} f(x-1)\,g(x)}$$

$$\Rightarrow e^{f(x-1)g(x)}$$

$$\therefore \lim_{x\to a} f(x)^{g(x)} = \lim_{x\to a} e^{f(x-1)g(x)}$$

if $\lim_{x\to a} f(x) \to 1$

$\lim_{x\to a} g(x) \to \infty$

Ques: $\lim_{x\to 0} (\cos x)^{1/x^2}$

Sol: $e^{(\cos x-1)\frac{1}{x^2}} \Rightarrow e^{\frac{-2\sin^2\frac{x}{2}}{x^2}}$

$$\Rightarrow e^{\frac{-2\sin^2\frac{x}{2}}{\frac{x^2}{4}\times 4}}$$

$$\Rightarrow \boxed{e^{-1/2}}$$

Ques: $\lim_{x\to \frac{\pi}{2}} (\sin x)^{\tan x}$

Sol: $e^{(\sin x-1)\tan x} \Rightarrow e^{0\times\infty} \Rightarrow e^0 = \boxed{1}$

Ques: $\lim_{x\to 0} (1+\ln(1+x))^{\frac{1}{x}}$

Sol: $e^{(1+\ln(1+x)-1)\frac{1}{x}} = e^{\frac{\ln(1+x)}{x}} = e^1 = \boxed{e}$

Ques: $\lim_{x\to\infty} \left(\frac{x+2014}{x+2015}\right)^{x+2016}$

Sol: $e^{\left(\frac{x+2014}{x+2015}-1\right)(x+2016)}$

$$= e^{-1\left(\frac{x+2016}{x+2015}\right)}$$

$$= e^{-1} = \boxed{\frac{1}{e}}$$

Ques: $\lim_{x\to\infty} \left(\frac{x^2+2x+3}{x^2+4x+7}\right)^{(3x+2)}$

Sol: $e^{\left(\frac{x^2+2x+3}{x^2+4x+7}-1\right)(3x+2)}$

$\Rightarrow e^{\left(\frac{x^2+2x+3-x^2-4x-7}{x^2+4x+7}\right)(3x+2)}$

$\Rightarrow e^{\frac{(-2x-4)}{x^2+4x+7} \times 3x+2} \Rightarrow e^{\frac{-2(x+2)(3x+2)}{x^2+4x+7}}$

$\Rightarrow e^{-2\left(\frac{3x^2+8x+4}{x^2+4x+7}\right)}$

$\Rightarrow \boxed{e^{-6} \quad \text{Ans.}}$

Ques: $\lim_{x \to \frac{\pi}{4}} (\tan x)^{\tan 2x}$

Sol: $(\tan x)^{\tan 2x} \Rightarrow e^{(\tan x - 1)\tan 2x}$

$= e^{(\tan x-1)\frac{2\tan x}{(1+\tan x)(1+\tan x)}}$

$= e^{\frac{-2\tan x}{1+\tan x}}$

$= e^{-1} = \boxed{\frac{1}{e} \quad \text{Ans.}}$

Ques: $\lim_{x \to 0} \left(\frac{a^x + b^x + c^x}{3}\right)^{\frac{2014}{x}}$

Sol: $e^{\left(\frac{a^x+b^x+c^x}{3}-1\right)\frac{2014}{x}} \Rightarrow e^{\left(\frac{a^x-1+b^x-1+c^x-1}{3}\right)\frac{2014}{x}}$

$\Rightarrow e^{\left(\frac{a^x-1}{x} + \frac{b^x-1}{x} + \frac{c^x-1}{x}\right)\frac{2014}{3}}$

$\Rightarrow e^{(\log a + \log b + \log c)\frac{2014}{3}}$

$\Rightarrow e^{\log(a+b+c)\frac{2014}{3}}$

$\Rightarrow \boxed{(abc)^{\frac{2014}{3}}}$

Ques: $\lim_{x \to \infty} \left(\frac{1^{\frac{1}{x}} + 2^{\frac{1}{x}} + \dots n^{\frac{1}{x}}}{n}\right)^{nx}$

Sol: $e^{\left(\frac{1^{\frac{1}{x}} + 2^{\frac{1}{x}} + \dots n^{\frac{1}{x}}}{n}\right) \times nx} \Rightarrow e^{(1^{\frac{1}{x}} + 2^{\frac{1}{x}} + \dots n^{\frac{1}{x}} - n)\times x}$

$\Rightarrow e^{\frac{(1^{\frac{1}{x}} + 2^{\frac{1}{x}} + \dots n^{\frac{1}{x}} - n)}{\frac{1}{x}}} \Rightarrow e^{\left(\frac{1^{\frac{1}{x}}-1}{1/x} + \frac{2^{1/x}-1}{1/x} + \dots \frac{n^{1/x}-1}{1/x} + n - n\right)}$

$\Rightarrow e^{(\ln 1 + \ln 2 + \ln 3 + \dots \ln n)}$

$\Rightarrow \boxed{n!}$

Ques: $\lim_{x \to 1} \left(\tan\left(\frac{\pi}{4} + \ln x\right)\right)^{1/\ln x}$

Sol: $e^{[\tan(\frac{\pi}{4} + \ln x) - 1]\frac{1}{\ln x}} = e^{\left(\frac{1+\tan \ln x}{1-\tan \ln x}\right)\frac{1}{\ln x}}$

$\Rightarrow e^{\left(\frac{2\tan \ln x}{1-\tan \ln x}\right) \times \frac{1}{\ln x}}$

$\Rightarrow e^{\left(\frac{2\sin \ln x}{\cos \ln x - \sin \ln x}\right) \times \frac{1}{\ln x}}$

$\Rightarrow e^{\frac{2\sin \ln x}{\ln x} \times \left(\frac{1}{\cos \ln x - \sin \ln x}\right)}$

$\Rightarrow e^{\frac{2}{\cos \ln x - \sin \ln x}}$

$\Rightarrow e^{\frac{2}{1-0}} = \boxed{e^2}$ Ans.

Ques: $\lim_{x \to 0} \left(\sin^2 \frac{\pi}{2-ax}\right)^{\sec^2\left(\frac{\pi}{2-bx}\right)}$

Sol: $e^{\left(\sin^2 \frac{\pi}{2-ax} - 1\right) \sec^2 \frac{\pi}{2-bx}}$

$\Rightarrow e^{-\cos^2 \frac{\pi}{2-ax} \times \sec^2 \frac{\pi}{2-bx}}$

$\Rightarrow e^{\left(-\cos^2 \frac{\pi}{2-ax} \,/\, \cos^2 \frac{\pi}{2-bx}\right)}$ → Solving separately

$$\frac{\sin^2\left(\frac{\pi}{2} - \frac{\pi}{2-ax}\right)}{\sin^2\left(\frac{\pi}{2} - \frac{\pi}{2-bx}\right)}$$

$$= \frac{\sin^2\left(\frac{\pi}{2} - \frac{\pi}{2-ax}\right) \times \left(\frac{\pi}{2} - \frac{\pi}{2-ax}\right)^2}{\left(\frac{\pi}{2} - \frac{\pi}{2-ax}\right)^2} \Bigg/ \frac{\sin\left(\frac{\pi}{2} - \frac{\pi}{2-bx}\right) \times \left(\frac{\pi}{2} - \frac{\pi}{2-bx}\right)^2}{\left(\frac{\pi}{2} - \frac{\pi}{2-bx}\right)^2}$$

$$= \left(\frac{\frac{\pi}{2} - \frac{\pi}{2-ax}}{\frac{\pi}{2} - \frac{\pi}{2-bx}}\right)^2$$

$= \frac{a + (2-bx)}{b(2-ax)} \Rightarrow \frac{2a}{2b} = \left(\frac{a}{b}\right)^2$

$= \boxed{e^{-a^2/b^2}}$

Ques: $\lim_{n \to \infty} \left(\frac{a-1+(b)^{1/n}}{a}\right)^n$

Sol: $e^{\left(\frac{a-1+b^{1/n}-1}{a}\right)n} \Rightarrow e^{\frac{(a-1+b^{1/n}-a)n}{a}}$

$\Rightarrow e^{(b^{1/n}-1)n/a} = e^{\frac{b^{(1/n-1)}}{1/n \times a}}$

$\Rightarrow e^{\ln b/a} \Rightarrow \boxed{b^{1/a}}$ Ans.

Ques: $\lim_{n\to\infty} \left(\frac{3}{n^3}\sum_{r=1}^{n} r^2\right)^n$

Sol: $\left[\frac{3}{n^3} \times \frac{n(n+1)(2n+1)}{6}\right]^n$

$\Rightarrow \left[\frac{3(n+1)(2n+1)}{6n^2}\right]^n = \left[\frac{2n^2+3n+1}{2n^2}\right]^n$

$\Rightarrow \left[1+\frac{3}{2n}+\frac{1}{2n^2}\right]^n$

$\Rightarrow e^{\left(1+\frac{3}{2n}+\frac{1}{2n^2}-1\right)n}$

$\Rightarrow e^{\left(n+\frac{3}{2}+\frac{1}{2n}-n\right)}$

$\Rightarrow \boxed{e^{3/2}}$ Ans.

Ques: $\lim_{x\to 0} \frac{\ln\left(\tan\left(\frac{\pi}{4}+ax\right)\right)}{\sin bx}$

Sol: $\frac{\ln\left(\tan\left(\frac{\pi}{4}+ax\right)\right)}{\frac{\sin bx}{bx} \times bx} \Rightarrow \frac{\ln\left(\tan\left(\frac{\pi}{4}+ax\right)\right)}{bx}$

$\Rightarrow \frac{\ln\left[\frac{1+\tan ax}{1-\tan ax}\right]}{bx}$

$\Rightarrow \ln\left[\frac{1+ax}{1-ax}\right] \times \frac{1}{bx}$

$\Rightarrow [\ln(1+ax) - \ln(1-ax)] \times \frac{1}{bx}$

$\Rightarrow \left[\frac{\ln(1+ax)\,ax}{ax} - \frac{(-ax)\ln(1+(-ax))}{-ax}\right] \times \frac{1}{bx}$

$\Rightarrow (ax+ax) \times \frac{1}{bx}$

$\Rightarrow \boxed{\frac{2a}{b}}$

Ques: If $f(x)$ is a polynomial with least degree which satisfy

$\lim_{x\to 0}\left(1+\frac{x^2+f(x)}{x^2}\right)^{1/x} = e^2$. Find $f(x)$.

Sol: $e^{\left(1+\frac{x^2+f(x)}{x^2}-1\right)\frac{1}{x}} \Rightarrow e^{\left(\frac{x^2+x^2+f(x)-x^2}{x^2}\right)\times\frac{1}{x}}$

$\Rightarrow e^{\left(\frac{x^2+f(x)}{x^2}\times\frac{1}{x}\right)} \Rightarrow e^{\frac{x^2+f(x)}{x^3}} \Rightarrow e^2$

$\frac{x^2+f(x)}{x^3} = 2 \Rightarrow x^2+f(x) = 2x^3$

$\boxed{f(x) = 2x^3 - x^2}$

L'HOSPITAL RULE

If $f(x)$ & $g(x)$ be two differentiable functions in the open interval containing 'a' except possibly at $x=a$ and $\lim_{x\to a} f(x) = 0, \pm\infty$, $\lim_{x\to a} g(x) \to 0, \pm\infty$ then $\lim_{x\to a}\frac{f(x)}{g(x)} \to \frac{0}{0}, \frac{\infty}{\infty}$ then we find the limit by differentiating the functions. That is, the limit is now $\lim_{x\to a}\frac{f'(x)}{g'(x)}$, if it is again inform of $\frac{0}{0}, \frac{\infty}{\infty}$ we again differentiate.

$x=a$, $\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ $\{f(a) = g(a) = 0\}$

$\frac{f'(a)}{g'(a)} = \lim_{h\to 0}\frac{\frac{f(a+h)-f(a)}{h}}{\frac{g(a+h)-g(a)}{h}} = \lim_{h\to 0}\frac{f(a+h)-f(a)}{g(a+h)-g(a)}$

$= \lim_{h\to 0}\frac{f(a+h)}{g(a+h)} \Rightarrow \lim_{x\to 0}\frac{f(x)}{g(x)}$

SANDWICH THEOREM

If $g(x) \le f(x) \le h(x)\ \forall x$ and if $\lim_{x\to 0} g(x) = \lim_{x\to 0} h(x) = l$ then 'l' is the limit of $f(x)$ at $x\to 0$

eg. $\lim_{x\to 0} \frac{\sin x}{x}$

$-1 \le \sin x \le 1$

$\frac{-1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$

$g(x) \quad f(x) \quad h(x)$

$\lim_{x\to\infty} h(x) = \lim_{x\to\infty} g(x) = 0$

$\lim_{x\to\infty} \frac{\sin x}{x} = 0$

Ques: $\lim_{x\to\infty} \frac{[x]}{x}$

Sol: $\frac{x - \{x\}}{x} = 1 - \frac{\{x\}}{x}$

$\{x\} \in [0,1)$

$\frac{\{x\}}{x} \in [0, \frac{1}{x})$

$-\frac{\{x\}}{x} \in (\frac{-1}{x}, 0]$

$1 - \frac{\{x\}}{x} \in (1 - \frac{1}{x}, 1]$

$\lim_{x\to\infty} g(x) = 1 = \lim_{x\to\infty} h(x) = 1$

$\lim_{x\to\infty} f(x) = 1$

Ques: $\lim_{x\to\infty} \frac{[x] + [2x] + \dots + [nx]}{n^2}$

Sol:

$x - 1 < [x] \le x$

$2x - 1 < [2x] \le 2x$

$\vdots \quad \vdots \quad \vdots$

$nx - 1 < [nx] \le nx$

$x\frac{n(n+1)}{2} - n < [x] + \dots [nx] \le x\frac{n(n+1)}{2}$

$\Rightarrow \frac{xn(n+1) - 2n}{2n^2} < \frac{[x] + [nx]}{n^2} \le \frac{xn(n+1)}{2n^2}$

$\Rightarrow \frac{x(n)(n+1)}{2n^2} - \frac{n}{2n^2} = \frac{x(n+1)}{2n} - \frac{1}{2n}$

$\Rightarrow \frac{x}{2}\left(1 + \frac{1}{n}\right) - \frac{1}{n}$

$\lim_{x\to\infty} f(x) = \frac{x}{2}$

Ques: $\lim_{n\to\infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+2} + \dots + \frac{n}{n^2+n}\right)$

Sol: $\frac{n}{n^2+1} + \frac{n}{n^2+2} + \dots + \frac{n}{n^2+n}$ } $f(x)$

$\Rightarrow \frac{n}{n^2+n} + \frac{n}{n^2+n} + \dots + \frac{n}{n^2+n} < f(x) < \frac{n}{n^2+1} + \frac{n}{n^2+1} + \dots + \frac{n}{n^2+1}$

$\Rightarrow \frac{n^2}{n^2+n} < f(x) < \frac{n^2}{n^2+1}$

$\Rightarrow \lim_{n\to\infty} \frac{n^2}{n^2+n} = \frac{1}{1+\frac{1}{n}} = 1$

$\Rightarrow \boxed{\lim_{n\to\infty} f(x) = 1}$

Ques: $\lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{\sqrt{n^n+k}} \right)$

Sol: $\frac{1}{\sqrt{n^2+n}} + \dots + \frac{1}{\sqrt{n^2+n}} < \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+n}} < \frac{1}{\sqrt{n^2+1}} + \dots \frac{1}{\sqrt{n^2+1}}$

$\Rightarrow \frac{n}{\sqrt{n^2+n}} < f(x) < \frac{n}{\sqrt{n^2+1}}$

$\lim_{n\to\infty} \frac{n}{\sqrt{n^2+1}} \Rightarrow \frac{n}{n\sqrt{1+\frac{1}{n^2}}} \Rightarrow \frac{1}{\sqrt{1+\frac{1}{n^2}}} = 1$

$\frac{n}{\sqrt{n^2+n}} \Rightarrow \frac{n}{n\sqrt{1+\frac{1}{n}}} \Rightarrow \frac{1}{\sqrt{1+\frac{1}{n}}} = 1$

$\boxed{\lim_{n\to\infty} f(x) = 1}$

Ques: $\lim_{x\to 2} \frac{(\cos\alpha)^x + (\sin\alpha)^x - \cos^2\alpha - \sin^2\alpha}{x-2}$

Sol: $\lim_{x\to 2} \cos^2\alpha \left(\frac{\cos^{x-2}\alpha - 1}{x-2}\right) + \sin^2\alpha \left(\frac{\sin^{x-2}\alpha - 1}{x-2}\right)$

$= \cos^2\alpha \ln\cos\alpha + \sin^2\alpha \ln\sin\alpha$

Ques: If $a_n = 2\cos\left(\frac{\theta}{2^n}\right) - 1$ then $\lim_{n\to\infty} (a_1 a_2 \dots a_n) = ?$

Sol: $a_1 = 2\cos\left(\frac{\theta}{2}\right) - 1$

$a_2 = 2\cos\left(\frac{\theta}{4}\right) - 1$

$\vdots$

$a_n = 2\cos\left(\frac{\theta}{2^n}\right) - 1$

$$a_1 \cdot a_2 \cdots a_n = \left[2\cos\left(\frac{\theta}{2}\right)-1\right]\left[2\cos\left(\frac{\theta}{4}\right)-1\right] \cdots \left[2\cos\left(\frac{\theta}{2^n}\right)-1\right]$$

$$= \frac{2\cos\left(\frac{\theta}{2^n}\right)-1}{2\cos\left(\frac{\theta}{2^n}\right)+1} \times 2\cos\left(\frac{\theta}{2^n}\right)+1$$

$$= \frac{4\cos^2\frac{\theta}{2^n}-1}{2\cos\frac{\theta}{2^n}+1}$$

$$= \frac{2\left(2\cos^2\frac{\theta}{2^n}\right)-1}{2\cos\frac{\theta}{2^n}+1} \Rightarrow \frac{2\left(1+\cos\frac{\theta}{2^{n-1}}\right)-1}{2\cos\frac{\theta}{2^n}+1}$$

$$= \frac{1+2\cos\frac{\theta}{2^{n-1}}}{2\cos\frac{\theta}{2^n}+1}$$

$$\Rightarrow \frac{\left[2\cos\left(\frac{\theta}{2}\right)-1\right]\cdots\left[2\cos\left(\frac{\theta}{2^{n-1}}\right)-1\right]\left[2\cos\left(\frac{\theta}{2^{n-1}}\right)+1\right]}{\left(2\cos\left(\frac{\theta}{2^n}\right)+1\right)}$$

$$\Rightarrow \frac{\left(2\cos\frac{\theta}{2}-1\right)\left(2\cos\frac{\theta}{4}-1\right)\left(2\cos\frac{\theta}{4}+1\right)}{2\cos\left(\frac{\theta}{2^n}\right)+1}$$

$$\Rightarrow \frac{\left[2\cos\left(\frac{\theta}{2}\right)-1\right]\left[2\cos\left(\frac{\theta}{2}\right)+1\right]}{2\cos\left(\frac{\theta}{2^n}\right)+1}$$

$$\Rightarrow \frac{2\left[1+\cos\theta\right]-1}{2\cos\left(\frac{\theta}{2^n}\right)+1}$$

$$\Rightarrow \frac{1+2\cos\theta}{2\cos\left(\frac{\theta}{2^n}\right)+1}$$

$$\Rightarrow \boxed{\frac{1+2\cos\theta}{3}} \quad (n\to\infty)$$

Ques: If $f(x) = \tan\frac{x}{2}\sec x + \tan\frac{x}{2^2}\sec\frac{x}{2} + \cdots$ n terms and $g(x) = \tan\frac{x}{2^n} + f(x)$ then find:

(i) $\lim_{x\to 0}\left(\frac{g(x)}{x}\right)^{1/x}$ (ii) $\lim_{x\to 0}\left(\frac{g(x)}{x}\right)^{1/x^2}$

Sol: $f(x) = \tan\frac{x}{2}\sec x + \ldots + \tan\frac{x}{2^n}\sec\frac{x}{2^{n-1}}$

$$\tan\frac{x}{2}\sec x = \frac{\sin\frac{x}{2}}{\cos x\cdot\cos\frac{x}{2}} = \frac{\sin\left(x-\frac{x}{2}\right)}{\cos\frac{x}{2}\cdot\cos x}$$

$$= \frac{\sin x\cos\frac{x}{2}}{\cos\frac{x}{2}\cdot\cos x} - \frac{\cos x\sin\frac{x}{2}}{\cos x\cdot\cos\frac{x}{2}} = \tan x - \tan\frac{x}{2}$$

$T_1 = \tan x - \tan\frac{x}{2}$

$T_2 = \tan\frac{x}{2} - \tan\frac{x}{4}$

$\vdots$

$T_n = \tan\frac{x}{2^{n-1}} - \tan\frac{x}{2^n}$

$g(x) = \tan x$

$f(x) = \tan x - \tan\frac{x}{2^n}$

(i) $\lim_{x\to 0}\left(\frac{g(x)}{x}\right)^{\frac{1}{x}} \Rightarrow \lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x}} \Rightarrow 1^{\infty}$

$e^{\left(\frac{\tan x - 1}{x}\right)\frac{1}{x}} \Rightarrow e^{\frac{\tan x - x}{x^2}}$

$\frac{\tan x - x}{x^2} = \frac{0}{0} \Rightarrow$ Differentiating, $\frac{\sec^2 x - 1}{2x}$

Again differentiating, $\frac{2\sec^2 x\cdot\tan x}{2} = \frac{\sec^2 x\tan x}{1}$

$\lim_{x\to 0}\frac{\sec^2 x\tan x}{1} = 0$

$\boxed{e^0 = 1}$

(ii) $\lim_{x\to 0}\left(\frac{g(x)}{x}\right)^{\frac{1}{x^2}} \Rightarrow e^{\left(\frac{\tan x}{x}-1\right)\frac{1}{x^2}}$

$\Rightarrow \left(\frac{\tan x - x}{x}\right)\times\frac{1}{x^2} \Rightarrow \frac{\tan x - x}{x^3}$

Differentiating, $\frac{\sec^2 x - 1}{3x^2}$

Again, $\frac{2\sec^2 x\tan x}{6x} = \frac{1}{3}\frac{\sec^2 x\tan x}{x}$

$\Rightarrow \frac{\sec^2 x}{3} = \frac{1}{3}$

$\Rightarrow \boxed{e^{1/3}}$

Ques: If $f(n,\theta) = \left(1-\tan^2\frac{\theta}{2}\right)\left(1-\tan^2\frac{\theta}{2^2}\right)\ldots\ldots$ n terms, then find $\lim_{n\to\infty} f(n,\theta)$

Sol:

$$\left(1-\tan^2\frac{\theta}{2}\right)\left(1-\tan^2\frac{\theta}{2^2}\right)\ldots\ldots\left(1-\tan^2\frac{\theta}{2^n}\right)$$

$$\Rightarrow \frac{\cos\theta}{\cos^2\frac{\theta}{2}} \times \frac{\cos\frac{\theta}{2}}{\cos^2\frac{\theta}{4}} \times \ldots\ldots \frac{\cos\frac{\theta}{2^{n-1}}}{\cos^2\frac{\theta}{2^n}}$$

$$\Rightarrow \frac{\cos\theta}{\left(\cos\frac{\theta}{2}\cdot\cos\frac{\theta}{4}\ldots\cos\frac{\theta}{2^n}\right)^2}$$

$$\Rightarrow \frac{(\cot\theta)2^n}{\cot\left(\frac{\theta}{2^n}\right)}$$

$$\Rightarrow \boxed{\theta\cdot\cot\theta}$$

Ques: $\lim_{n\to\infty} \frac{1\cdot3\cdot5\ldots\ldots(2n-1)}{2\cdot4\cdot8\ldots2^n}$

Sol:

$$\frac{1}{2\cdot4\cdot8\ldots2^n} < \frac{1\cdot3\cdot5\ldots(2n-1)}{2\cdot4\cdot8\ldots2^n} < \frac{n^n}{2\cdot4\ldots2^n}$$

$$\Rightarrow \frac{1+3+5\ldots(2n-1)}{n} \geq [1\cdot3\cdot5\ldots(2n-1)]^{1/n}$$

$$\Rightarrow \frac{n}{2n}[2+(n-1)2] \geq (1\cdot3\cdot5\ldots(2n-1))^{1/n}$$

$$\Rightarrow n \geq [1\cdot3\cdot5\cdot(2n-1)]^{1/n}$$

$$\Rightarrow n^n \geq [1\cdot3\cdot5\ldots(2n-1)]$$

$\lim_{n\to\infty} \frac{1}{2\cdot4\cdot8\ldots2^n} \Rightarrow \frac{1}{\infty} \Rightarrow \boxed{0}$ Ans.

LEIBLITZ ROLE OF DIFFERENTIATION

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)\,dt = f[h(x)h'(x)] - f[g(x)g'(x)]$$

Ques: $\lim_{x\to0} \frac{\int_x^{x^2} 2t\,dt}{x^4}$

Sol: $\int_x^{x^2} 2t\cdot dt = 2x^2\times2x - 2x \Rightarrow 4x^3 - 2x$

$$\frac{4x^3-2x}{x^4} = \frac{12x^2-2}{4x^3} = -\frac{2}{0}$$

Limit $= -\infty$

Ques: $\lim_{x\to 0} \dfrac{\int_0^x \frac{t^2\,dt}{\sqrt{a+t}}}{bx-\sin x} = 1$, then find a & b

Sol: $\int_0^x \frac{t^2\,dt}{\sqrt{a+t}} = \frac{x^2}{\sqrt{a+x}} \times 1 - \frac{0}{\sqrt{a}}(0)$

$$= \frac{x^2}{\sqrt{a+x}}$$

$$\Rightarrow \frac{x^2}{\sqrt{a+x}\,(bx-\sin x)} = 1$$

$$\Rightarrow \frac{\frac{x^2}{\sqrt{a+x}}}{bx-\sin x} \Rightarrow \frac{\frac{\sqrt{a+x}\,2x - x^2 \times \frac{1}{2\sqrt{a+x}}}{a+x}}{b-\cos x}$$

$\Rightarrow b-\cos x = 0$

$x \to 0$, $\boxed{b=1}$

$$\lim_{x\to 0} \frac{\frac{x^2}{\sqrt{a+x}}}{b-\cos x} = 1 \Rightarrow \frac{\frac{x^2}{\sqrt{a+x}}}{1-\cos x} = 1$$

$$\Rightarrow \frac{x^2}{\sqrt{a+x} \times 2\sin^2\frac{x}{2}} = \frac{x^2}{\sqrt{a+x} \times 2 \times \frac{\sin^2\frac{x}{2}}{(x/2)^2} \times \left(\frac{x}{2}\right)^2} = 1$$

$$\Rightarrow \frac{4}{2\sqrt{a+x}} = 1 \Rightarrow \frac{2}{\sqrt{a}} = 1$$

$\Rightarrow \boxed{a=4}$

LIMITS IN MATRICS & DETERMINANTS

If $A(x) = \begin{bmatrix} f_1(x) & f_2(x) \\ f_3(x) & f_4(x) \end{bmatrix}$

$$\lim_{x\to a} A(x) = \lim_{x\to a} \begin{bmatrix} f_1(x) & f_2(x) \\ f_3(x) & f_4(x) \end{bmatrix} = \begin{bmatrix} \lim_{x\to a} f_1(x) & \lim_{x\to a} f_2(x) \\ \lim_{x\to a} f_3(x) & \lim_{x\to a} f_4(x) \end{bmatrix}$$

If any of the limits do not exist, limit of $A(x)$ does not exists.

If $\Delta(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_3(x) & f_4(x) \end{vmatrix}$

$\lim_{x \to a} \Delta(x) = \begin{vmatrix} \lim_{x \to a} f_1(x) & \lim_{x \to a} f_2(x) \\ \lim_{x \to a} f_3(x) & \lim_{x \to a} f_4(x) \end{vmatrix}$

$= \begin{vmatrix} l_1 & l_2 \\ l_3 & l_4 \end{vmatrix} = l_1 l_4 - l_2 l_3$

Ques: $\lim_{x \to 0} \dfrac{\ln(\ln(1-x^2))}{\ln(\ln(\cos x))}$

Sol:
$$\frac{\frac{1}{\ln(1-x^2)} \times \frac{1}{1-x^2} \times -2x}{\frac{1}{\ln(\cos x)} \times \frac{1}{\cos x} \times -\sin x}$$

$$\Rightarrow \frac{2\ln(\cos x)\cos x}{\ln(1-x^2)\,1-x^2}$$

$$\Rightarrow \frac{2\ln(1+(\cos x-1))\cos x}{(\cos x-1)\ln(1-x^2)\,1-x^2} \times (\cos x-1)$$

$$\Rightarrow \frac{2\cos x(\cos x-1)}{\ln(1-x^2)(1-x^2)}$$

$$\Rightarrow \frac{2\cos(\cos x-1)}{\frac{\ln(1+x^2)}{(-x^2)} \times -x^2(1-x^2)}$$

$$\Rightarrow \frac{2\cos x(\cos x-1)}{-x^2(1-x^2)}$$

$$\Rightarrow \frac{-2\cos x \times 2\sin^2\frac{x}{2}}{-\frac{x^2}{4} \times 4(1-x^2)}$$

$$\Rightarrow \frac{2\cos x \times 2}{4(1-x^2)}$$

$$\Rightarrow \boxed{1}$$